



University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

BMI/STAT-768

Statistical Methods
for

Medical Image Analysis

and High Dimensional
Data

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<https://github.com/lapcebeltrami/BMI768>

Medical Imaging Data Types

Unit Objectives

Understand different imaging data types

- 1) Identify imaging data types
- 2) Load images to MATLAB
- 3) Visualization and rendering
- 4) Get familiar with data and underlying science

Different Imaging data types

1) 1D time series, functional signal
-biological signal (EEG)

2) 2D volume images (microscope images)

3) 3D volume images (CT, PET, MRI)

4) Static network data (EEG, fMRI)

5) 2D surface data (obtained from 3D volume)

6) Dynamic imaging data

(additional temporal & longitudinal components)

7) Dynamic graph/network data

8) Directed dynamic networks

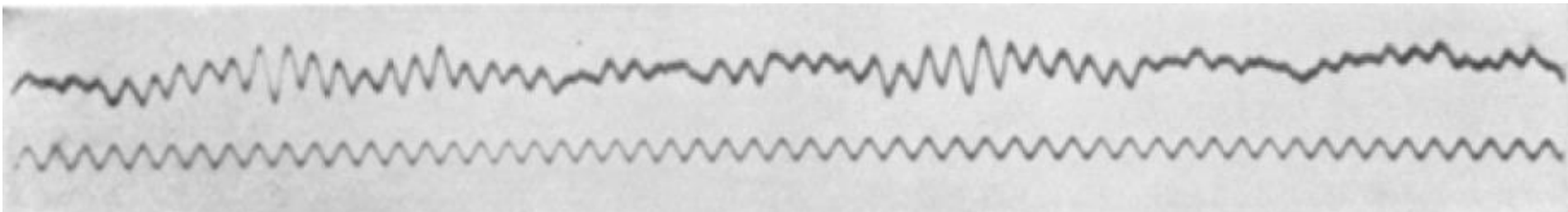
Basics:
1/3 of
lectures

Advanced:
2/3 of
lectures

Methods covered in the course

- 1) Matlab programming
- 2) Hypothesis testing, permutation test
- 3) Unsupervised Learning (clustering) is Statistical Inference
- 4) Green's function as probability distribution on manifolds
- 5) Scalable Wasserstein distance – optimal transport
- 6) Robust & sensitivity analysis, bounding, inequality
- 7) Limitation of causal models – Granger causality, SEM
- 8) Limitation of wavelets – diffusion wavelets, heat kernel
- 9) **Functional data analysis** - function as data
- 10) **Geometric data analysis** – geodesics, exponential map
- 11) **Topological data analysis** – data dimension, basis
- 10) **Dynamic data analysis** – temporal dimension
- 11) **Network data analysis** – dynamics, diffusion dynamics

Electroencephalography (EEG)



The first human EEG recording obtained by Hans Berger in 1924.

Required reading for EEG analysis and topological data analysis (TDA)

Wang et al. 2018 Annals of Applied Statistics

El-Yaagoubi et al. 2023 Entropy

ELECTROENCEPHALOGRAPHY (EEG)

1. WHAT IS EEG?

EEG is the noninvasive recording of **electrical activity** along the scalp.

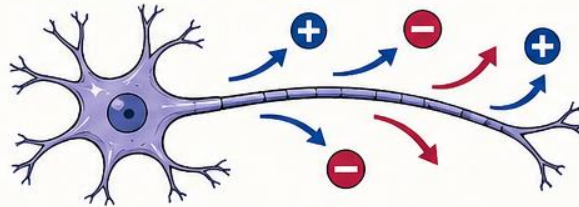


Neuronal activity generates tiny electrical signals.

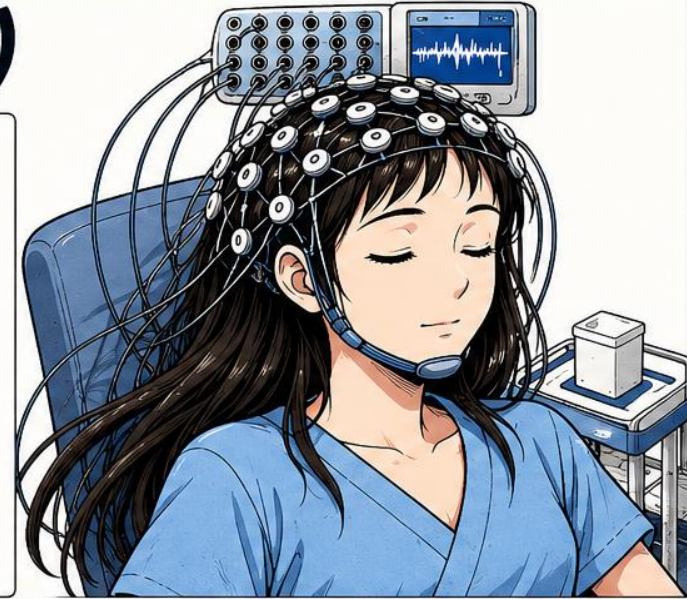
EEG measures **voltage fluctuations** over time.

2. WHAT DOES EEG SHOW?

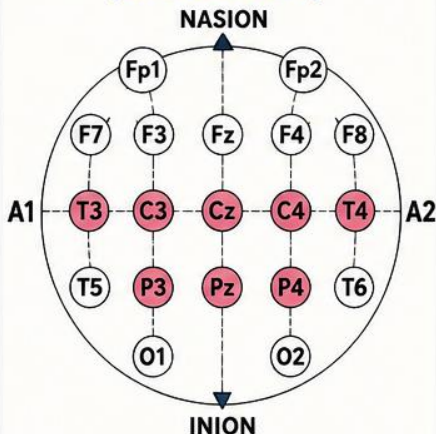
EEG reflects **ionic current flows** within the neurons of the brain.



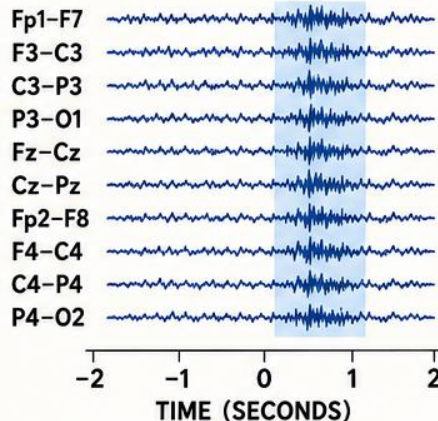
EEG does not measure individual neurons. It captures the **summed activity** of millions of neurons.



3. ELECTRODE PLACEMENT (10-20 SYSTEM)

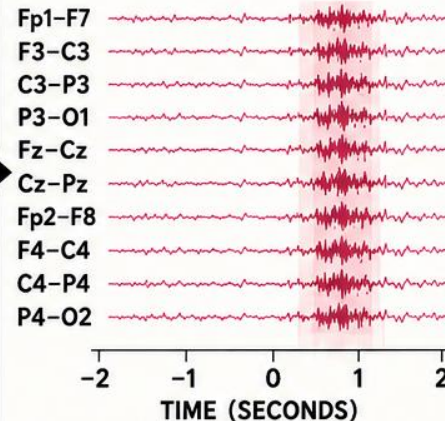


4. RAW EEG SIGNALS (TIME DOMAIN)



Each line is one channel. Signals vary over time across different brain regions.

5. WAVELET TRANSFORM (TIME-FREQUENCY)



Wavelet transform shows how frequency content changes over time in each channel.

6. KEY POINTS



EEG records **electrical activity** from the scalp.



Reflects the **summed activity** of many neurons.



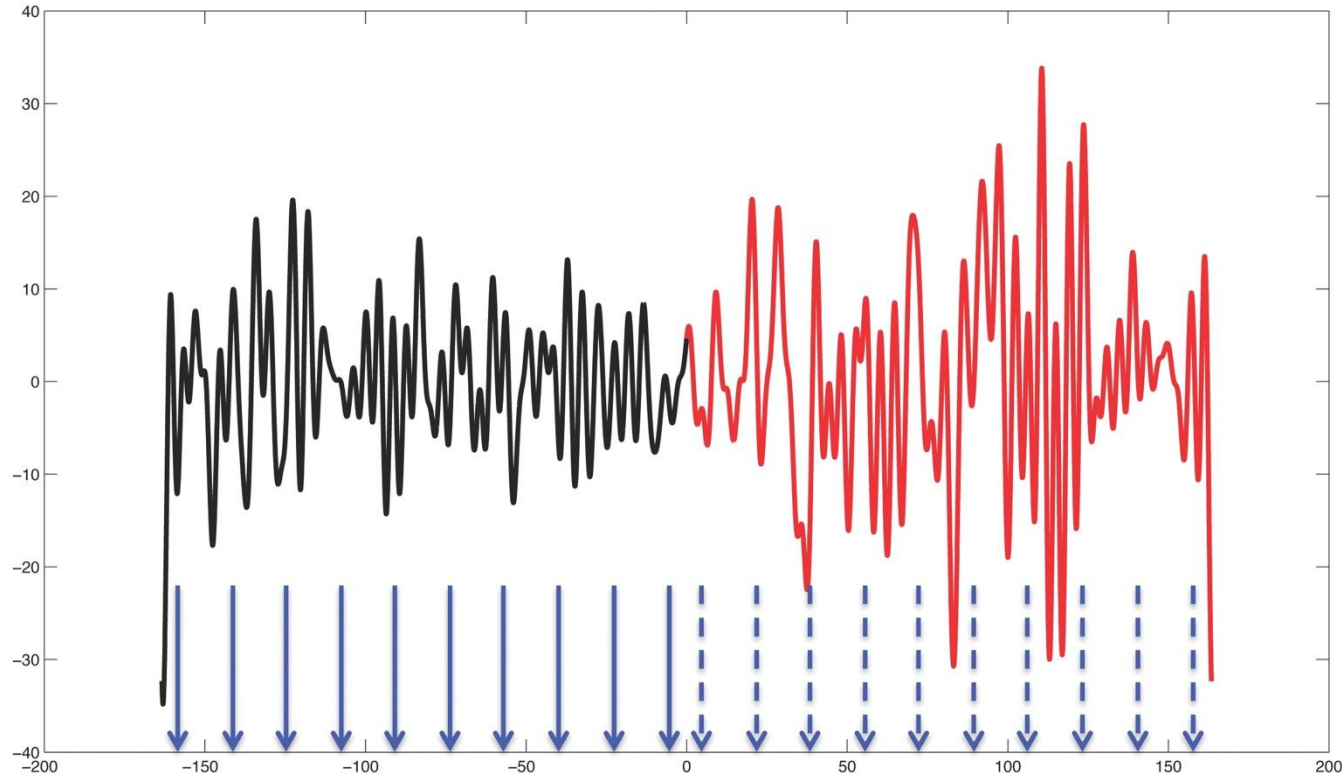
High temporal resolution (**milliseconds**).



Safe, painless, and widely used in clinical and research settings.

EEG in a single channel during seizure

Seizure video



Black = pre-seizure Red = seizure attack

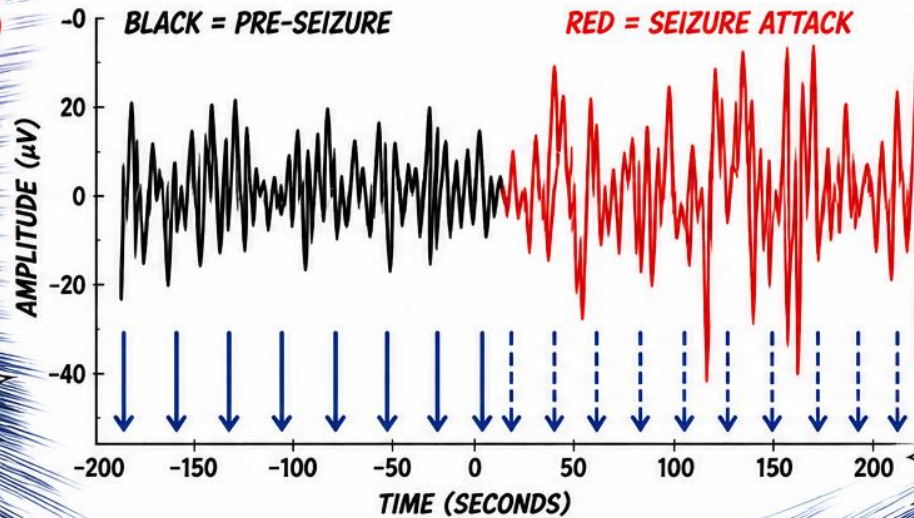
Wang et al. 2015 Topological epilepsy seizure detection in electroencephalographic signals, ISBI 351-354

THE PROBLEM IN EPILEPSY: DETECTING SEIZURES

EPILEPTIC SEIZURES CAN STRIKE **SUDDENLY** AND **UNPREDICTABLY!**

HOW CAN WE DETECT THEM IN TIME?

EEG SIGNAL: PRE-SEIZURE vs SEIZURE ATTACK



THE BRAIN SUDDENLY GOES INTO A **CHAOTIC STATE!**

THAT'S A **SEIZURE ATTACK!**

1 WHAT DOES THIS MEAN?

Each line is the brain's electrical activity at one electrode.



DURING A SEIZURE, THE BRAIN PRODUCES LARGE, IRREGULAR ELECTRICAL DISCHARGES.

SO THE SIGNAL BECOMES **STRONGER AND MORE CHAOTIC!**

2 WHY IS THIS IMPORTANT?



UNDETECTED SEIZURES CAN LEAD TO:

- **INJURY**
- **LOSS OF AWARENESS**
- **SUDEP (SUDDEN UNEXPECTED DEATH IN EPILEPSY)**
- **POOR QUALITY OF LIFE**



EARLY DETECTION CAN **SAVE LIVES!**

3 THE CHALLENGE

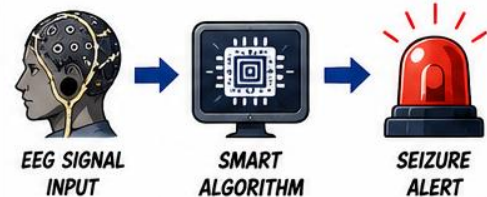
Seizures vary from person to person. They can start suddenly, last different amounts of time, and look different in the EEG.



CAN WE DETECT THE SWITCH **BEFORE** THE SEIZURE FULLY HAPPENS?

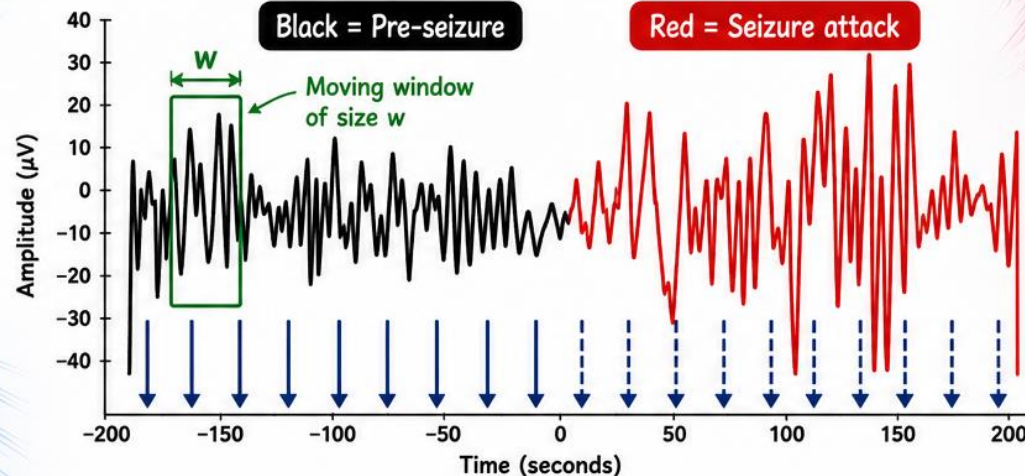
4 THE GOAL

Develop methods that can automatically analyze EEG signals and detect seizures as early and accurately as possible.



BETTER DETECTION MEANS **BETTER CARE** AND A **SAFER LIFE** FOR PEOPLE WITH EPILEPSY!

QUESTION:
COMPUTE THE
**MOVING
AVERAGE!**



WHAT IS THIS DATA
TELLING US?

1) EEG TIME SERIES DATA

We observe EEG signals
at discrete time points.

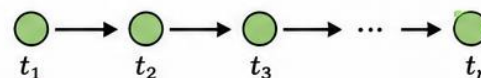
EEG values
 $\{f(t_1), f(t_2), \dots, f(t_n)\}$

Time points
 $t_1, t_2, \dots, t_n$

A **time series** is a sequence of
data points measured over time
at equal time intervals.

2) CAUSALITY MATTERS!

In time series, the order of time
carries important information.



The past can influence the future,
but the future cannot influence
the past.

Knowing the past
helps us understand
and predict
the future!

KEY IDEA

EEG signals are time series.
Causality is an important
characteristic.

So we should use
methods that respect
direction in time!

FUTURE PREDICTION



Can we predict
the time of
epilepsy seizure?

PAST (OBSERVED)



FUTURE (TO BE PREDICTED)



Goal: Predict when the seizure will occur
so we can warn and protect patients!

**EARLY
PREDICTION
CAN SAVE
LIVES!**



EEG dataset

Magnetic Resonance Imaging (MRI)



WHAT IS MRI? MAGNETIC RESONANCE IMAGING

USING MAGNETIC FIELDS AND RADIO WAVES TO SEE INSIDE THE BODY

MRI uses strong magnetic fields and radio waves to produce detailed images of the inside of the body without surgery or ionizing radiation.

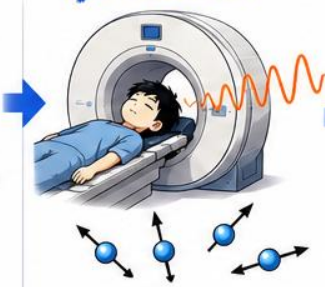


1 STRONG MAGNETIC FIELD (B_0)



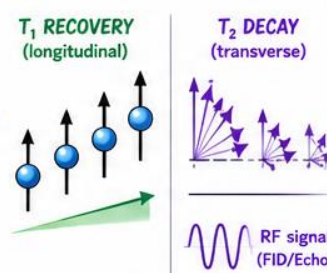
A strong magnetic field aligns the hydrogen protons in the body.

2 RADIOFREQUENCY PULSES (RF)



RF pulses at the Larmor frequency tip the aligned protons away from B_0 .

3 RELAXATION AND SIGNAL EMISSION



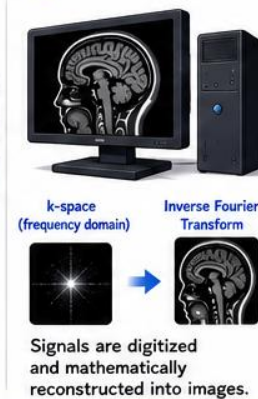
Protons relax (T_1 , T_2) and emit RF signals.

4 SIGNAL DETECTION



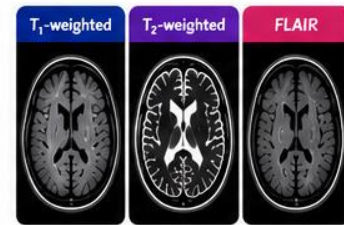
Receiver coils detect the tiny RF signals emitted by the relaxing protons.

5 COMPUTER PROCESSING



Signals are digitized and mathematically reconstructed into images.

6 MRI IMAGE OUTPUT



High-resolution images with excellent soft tissue contrast are produced.

MRI FIELD STRENGTH (CLINICAL SYSTEMS)

1.5 T
~2 mm



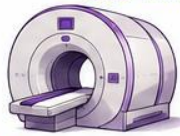
- ✓ Standard clinical use
- ✓ Good balance of signal and speed

3 T
~1 mm



- ✓ More signal
- ✓ Higher resolution
- ✓ Better contrast

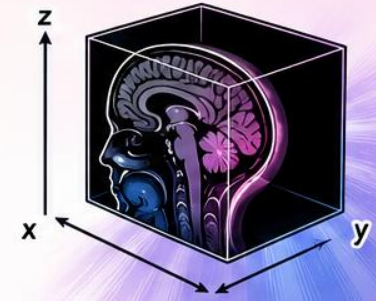
7 T
~0.5 mm



- ✓ Ultra-high resolution
- ✓ Research use
- ✓ More sensitive to susceptibility

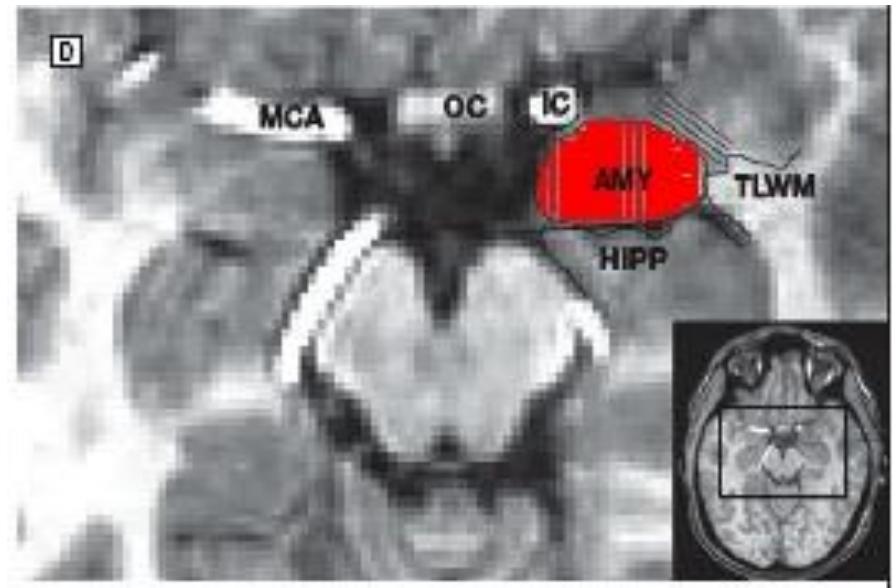
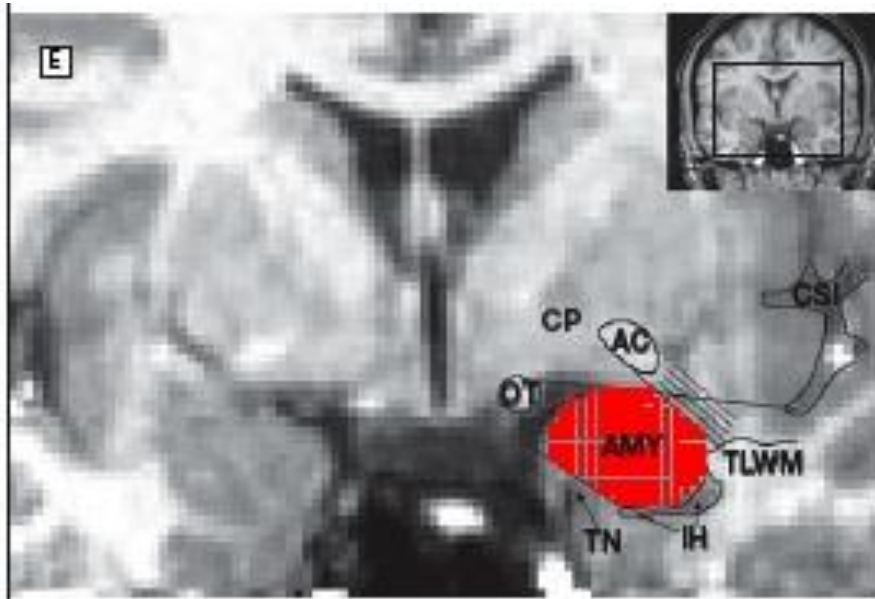
IN SUMMARY

- ✓ Strong magnetic field aligns protons.
- ✓ RF pulses tip protons away from alignment.
- ✓ Relaxation (T_1 , T_2) emits tiny RF signals.
- ✓ Coils detect the signals.
- ✓ Computers reconstruct high-resolution images.
- ✓ Higher field strength (1.5 T $\rightarrow$ 3 T $\rightarrow$ 7 T) $\rightarrow$ higher SNR and resolution.

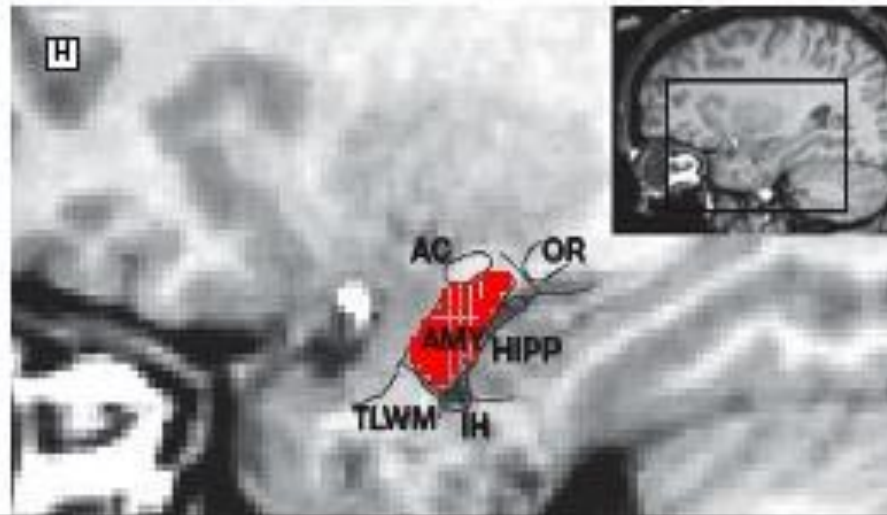


MRI PROVIDES EXCELLENT SOFT TISSUE CONTRAST AND IS WIDELY USED FOR BRAIN, SPINAL CORD, MUSCLES, JOINTS, AND MANY OTHER ORGANS!

3D volumetric data - image intensity value stored in 3D array



amygdala segmentation



Chung et al. 2010, NeuroImage

ROI-BASED VOLUMETRY: MEASURE VOLUME OF A BRAIN REGION!



We want to know:
Is the **amygdala volume** different
between **autism** and **control**?

To do that, we need
ROI-based volumetry!
Let's see how it works!



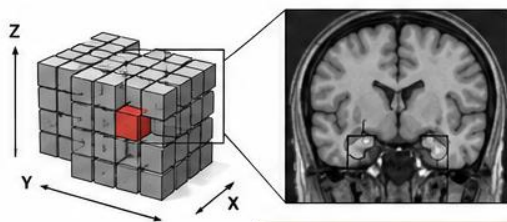
ROI (Region of Interest)
is a specific brain region
we want to measure.

We will use MRI
3D data and
segment (extract)
the **amygdala**.



1 3D VOLUMETRIC MRI DATA

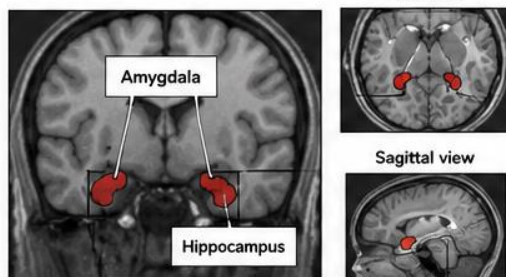
MRI gives us 3D images.
Each tiny cube (voxel) has an intensity value.



This 3D array of intensity values
is our **raw data**.

2 DEFINE ROI: THE AMYGDALA

The amygdala is a small structure deep in the
medial temporal lobe.

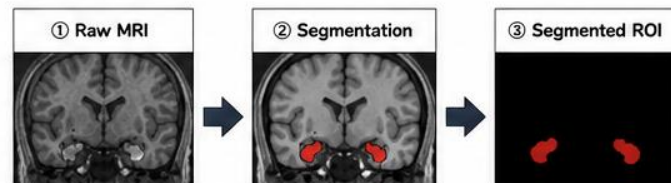


Axial view

Sagittal view

3 AMYGDALA SEGMENTATION

We automatically (or manually) segment the **amygdala**
from the 3D MRI.



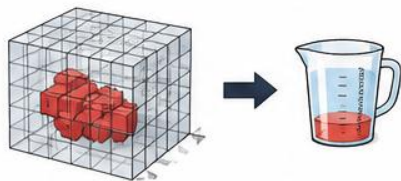
The red part is the
amygdala (our ROI)!



Now we have a clean 3D mask
of the amygdala.

4 COMPUTE VOLUME

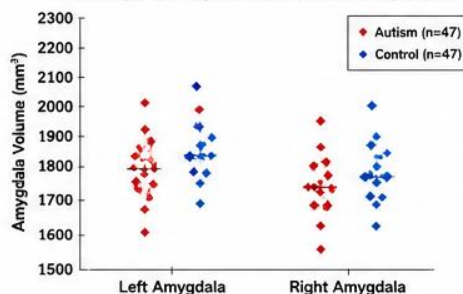
Volume = (number of voxels in ROI) × (voxel size)



- ✓ We count how many voxels are in the amygdala.
- ✓ Multiply by voxel size to get volume (mm³).

5 GROUP COMPARISON

We compare amygdala volume between groups.



No significant
volume difference
in autism vs.
control (n=47).
Left p=0.64
Right p=0.81



6 WHY THIS MATTERS

ROI-based volumetry gives us an objective and reliable
way to study brain structure.

- Helps understand neurodevelopment
in autism and other disorders.
- Provides quantitative data for
research and clinical studies.
- Can be combined with other measures
(e.g., shape, thickness, connectivity).
- Reproducible, objective, and
scalable to large datasets.

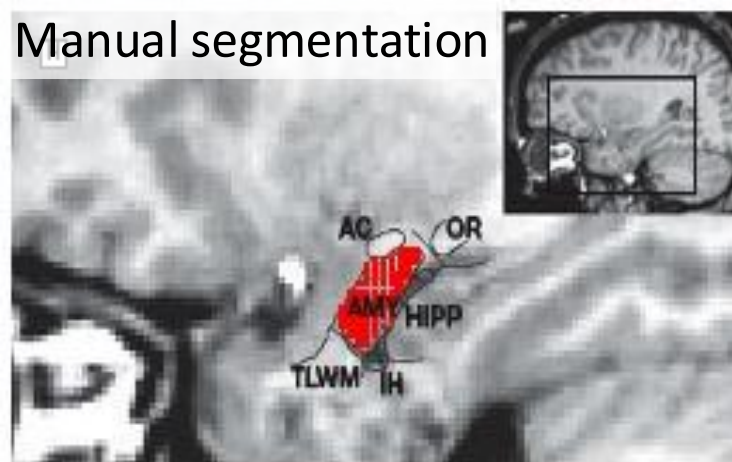


From 3D MRI
to meaningful
numbers!
**Science in
action!**



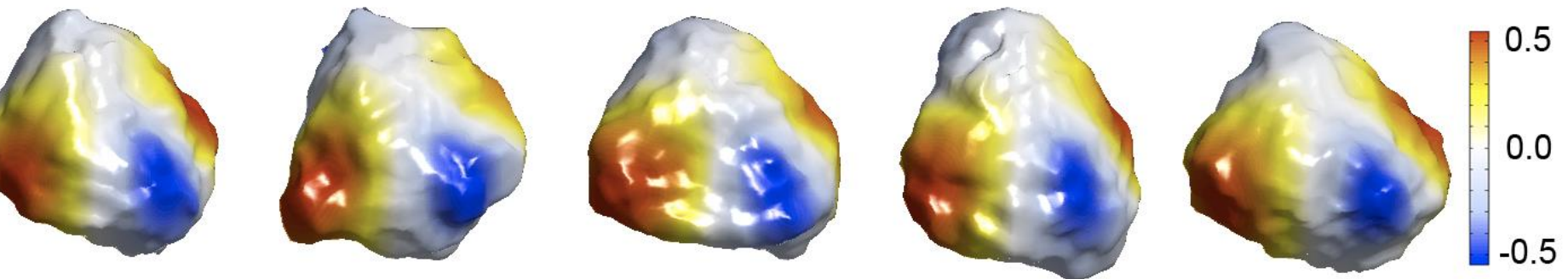
SUMMARY: ROI-BASED VOLUMETRY = 3D DATA → SEGMENTATION → VOLUME → STATISTICAL COMPARISON
A POWERFUL PIPELINE TO REVEAL HOW BRAIN STRUCTURE RELATES TO BEHAVIOR AND HEALTH!

Manual segmentation

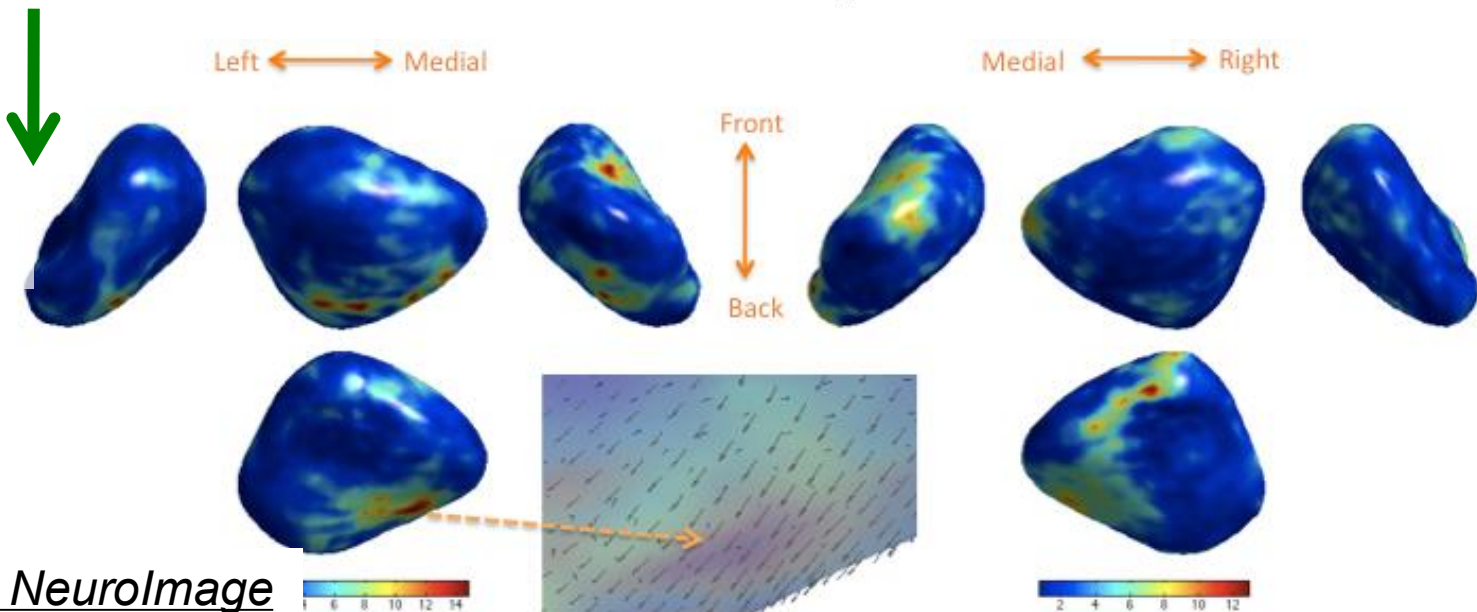


Geometric Data Analysis

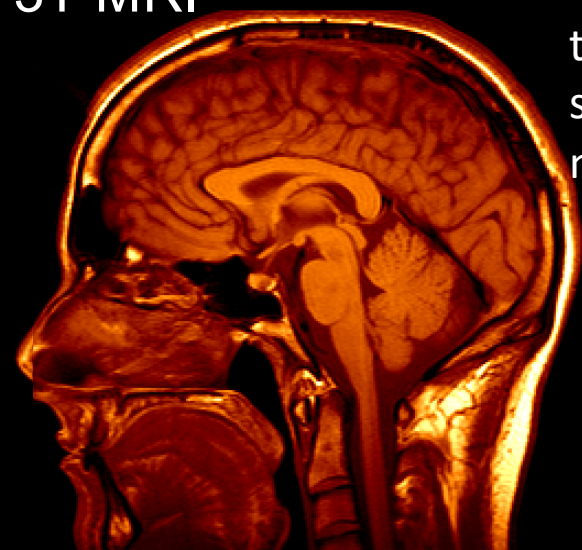
Surface registration
(colors show correspondence)



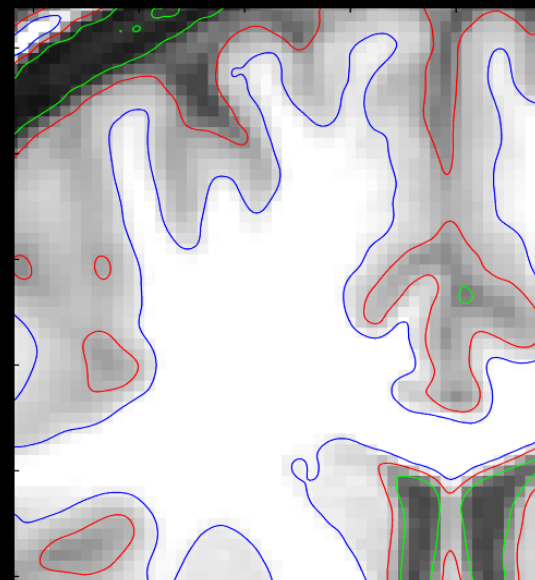
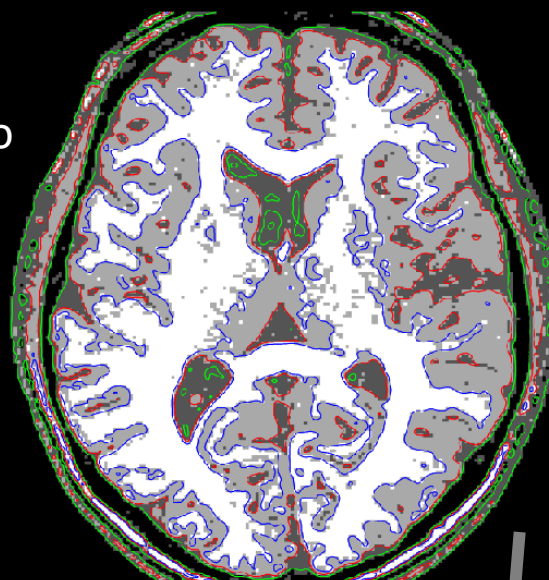
F-statistic map:
Autism vs. control
group difference



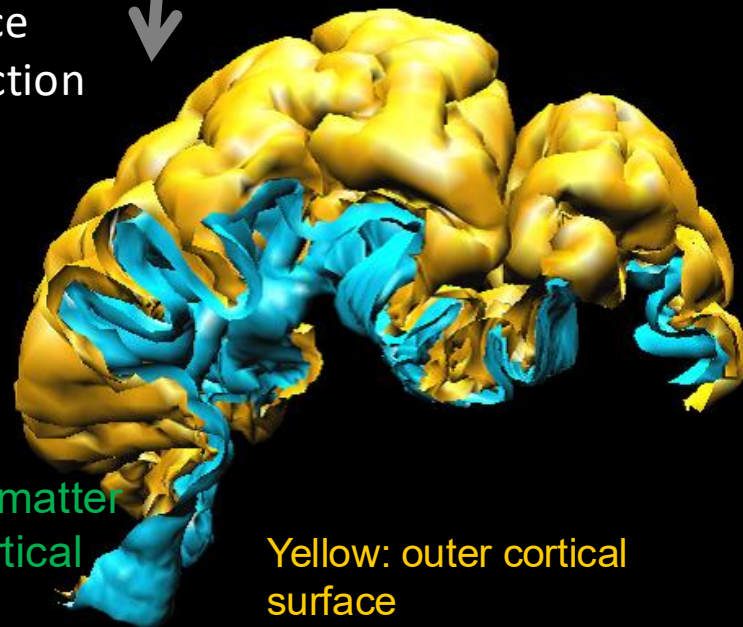
3T MRI



tissue
segmentation



surface
extraction



Smaller sulci
→ More gray matter
→ Thicker cortical
thickness

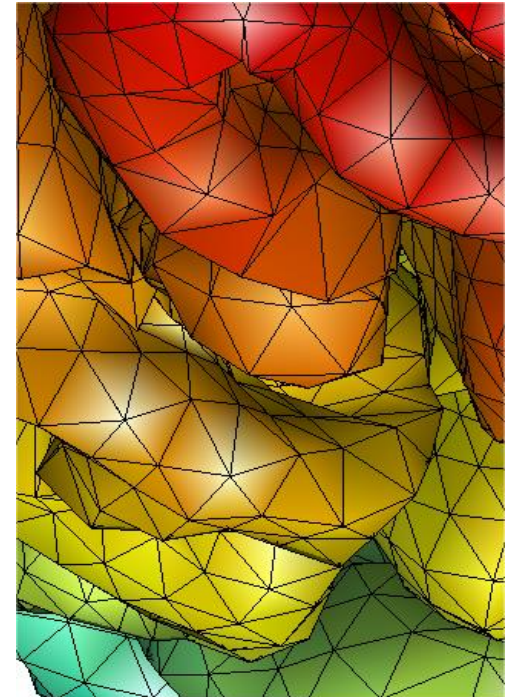
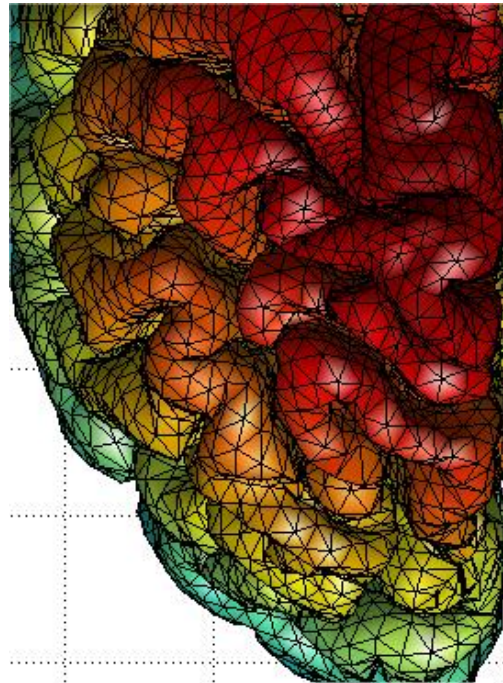
Yellow: outer cortical
surface
(gray matter boundary)
Blue: inner cortical surface
(white/gray matter
interface)



Triangle mesh with about 0.6 million
triangles per whole brain

Triangle mesh data

Basis of most surface rendering tools for 3D computer games:
as 3D Max Studio, Maya



*This is beyond current AI's capability
All they can do now is generating 1D images*

THE CORE STATISTICAL ISSUE: **NOISE** IN SURFACE DATA

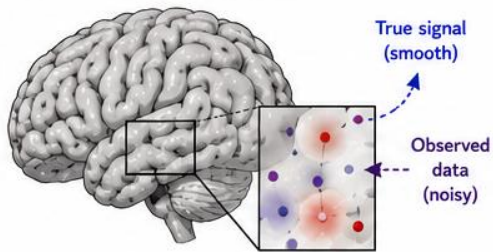
Brain surface measurements from MRI are inherently **noisy**.
Noise can **hide** true effects, **create** false ones,
and make results unreliable.

We can't just
look at raw data.
Noise is the biggest
obstacle to seeing
what is **real**!



1. WHAT IS NOISE?

Many factors add unwanted variability
to surface measurements.



Sources of noise include:

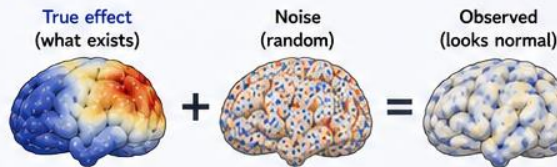
-  Scanner limitations (SNR, resolution)
-  Head motion and positioning
-  Physiological variability
-  Individual anatomical differences
-  Surface reconstruction and mapping

2. WHY NOISE IS A PROBLEM

Noise can **mislead** us in two dangerous ways.

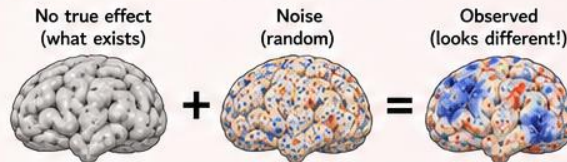
Mask true effects (False Negative)

Real, meaningful differences are
covered up by noise.



Create false effects (False Positive)

Random noise can look like real differences.



Without proper handling, we can be easily **misled**.

3. THE GOAL: SEE WHAT IS REAL

Use appropriate modeling and statistics to...



Reduce noise

Remove random fluctuations.



Preserve true signals

Keep meaningful patterns.



Respect the surface

Use methods that honor the
geometry and smoothness.



Make reliable inferences

Draw valid conclusions with
controlled uncertainty.



When we handle noise properly,
we reveal the **real** story hidden
in the brain's surface.



NOISE IS INEVITABLE. IGNORING IT IS NOT AN OPTION. HANDLE IT RIGHT—DISCOVER WHAT IS REAL.

Required reading for surface data analysis

Chung et al. arXiv:2203.06665

Computed Tomography (CT)



UW-Madison Radiology dept.



Let's see how we get a 3D image of lung blood vessels from a CT scan!

FROM CT SCAN TO 3D IMAGE!

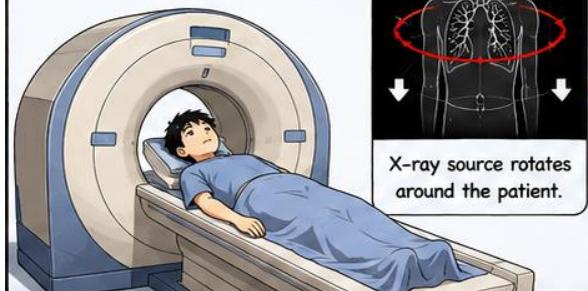
Visualizing lung blood vessels in 3D

It's like reconstructing the hidden roadmap inside the lungs!



1 CT SCANNING

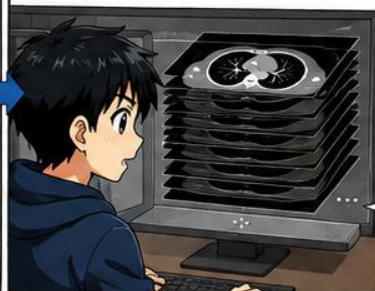
The patient lies in the CT scanner. X-rays are taken from many angles.



X-ray source rotates around the patient.

2 CT SLICES (CROSS-SECTIONS)

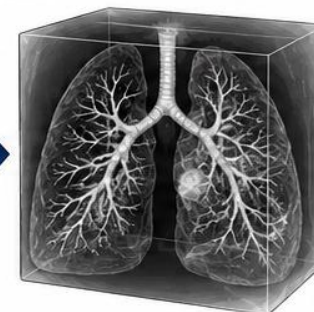
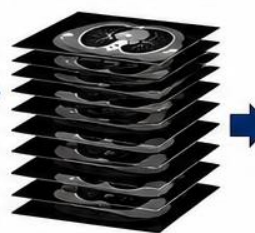
The computer rebuilds the data into many thin cross-sectional images.



Each slice shows a thin layer of the body.

3 VOLUME RECONSTRUCTION

All slices are combined to form a 3D volume of the lungs.



Now we have a 3D block containing all the information!

4 VESSEL ENHANCEMENT & SEGMENTATION

We enhance the blood vessels and separate them from other tissues.

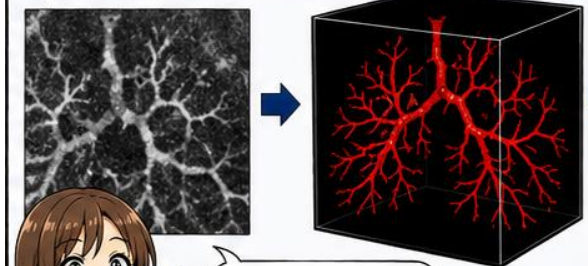
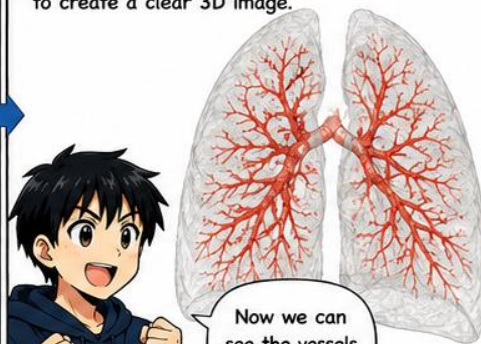


Image processing makes the vessels stand out and removes the noise!

5 3D RENDERING

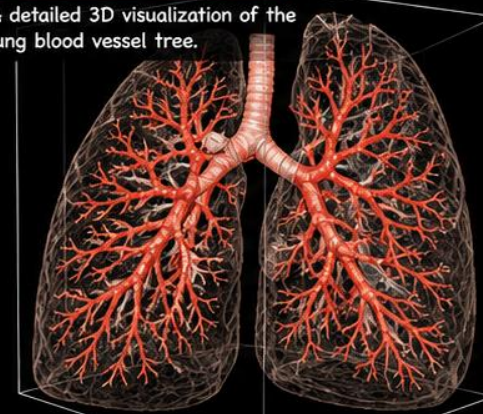
The segmented vessels are rendered to create a clear 3D image.



Now we can see the vessels in 3D!

6 FINAL 3D IMAGE

A detailed 3D visualization of the lung blood vessel tree.

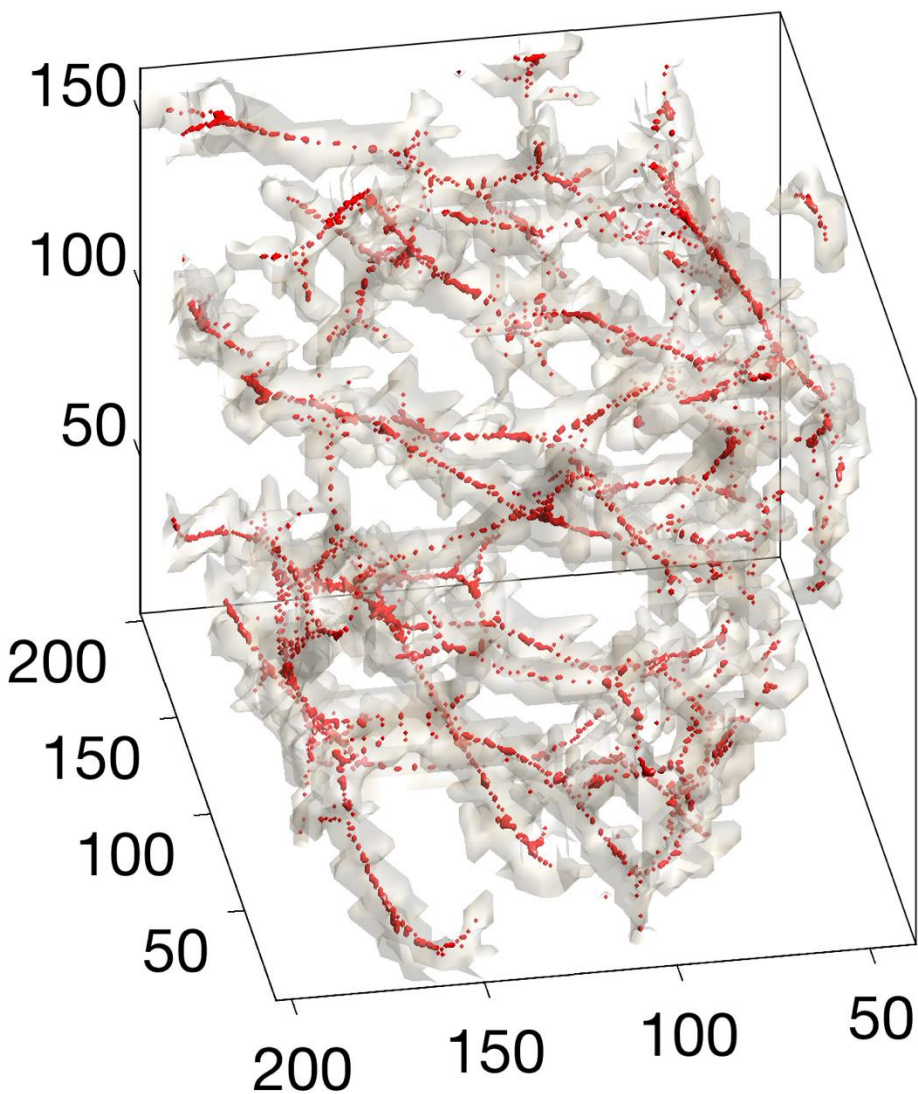


This helps doctors understand structure, detect problems, and plan treatment!

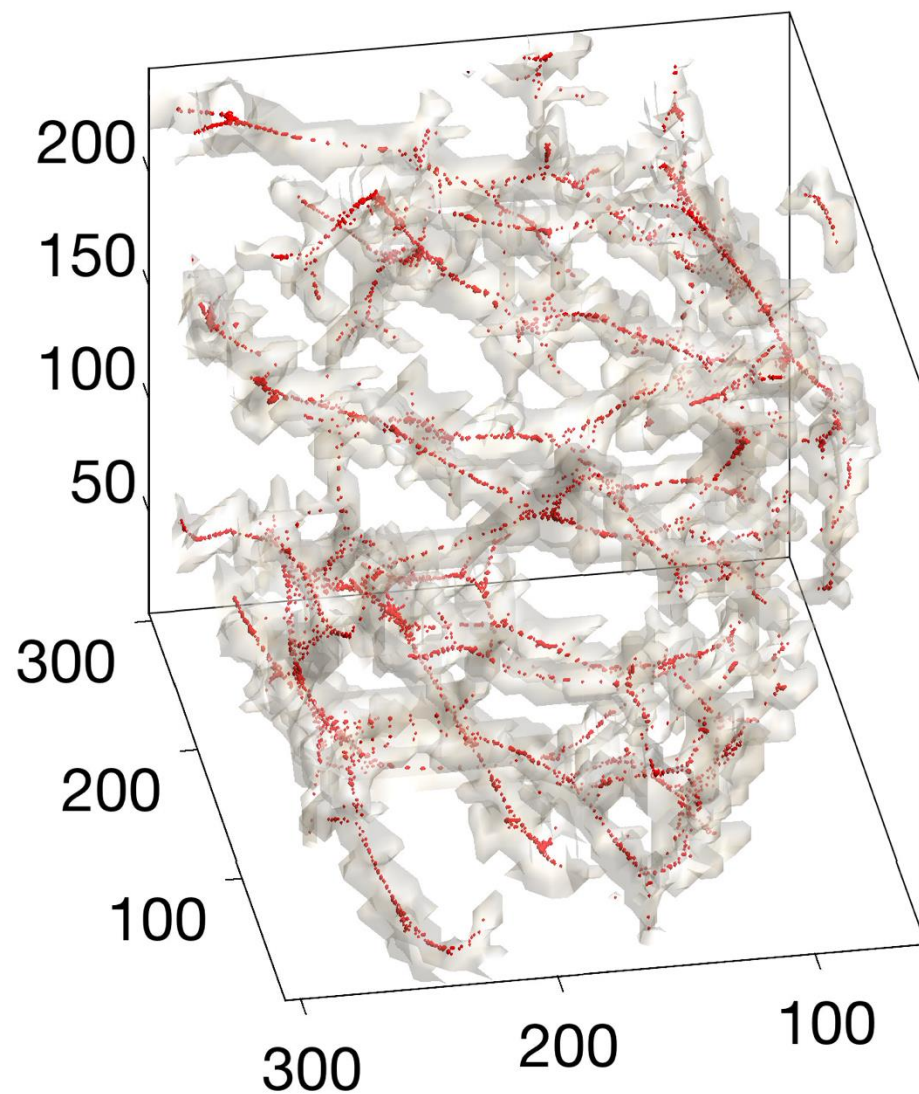


From many X-ray measurements → cross-sectional images → 3D volume → enhanced vessels → beautiful 3D visualization!

Tree Data Structure



Lung blood vessel trees from CT



Chung et al. 2018 EMBC

Chung et al. 2019, Mathematics of Shapes and

FROM 3D CT TO VESSEL TREE GRAPH!

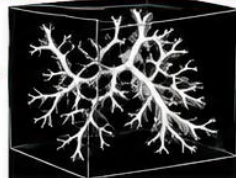
How we transform complex lung vessels into a simple graph.

1. 3D CT SCAN



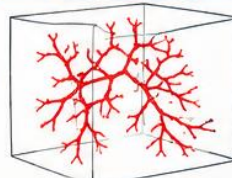
We start with a 3D CT image of the lungs.

2. VESSEL ENHANCEMENT & SEGMENTATION



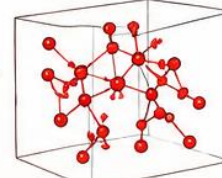
Enhance and segment the blood vessels.

3. SKELETONIZATION (MEDIAL AXIS)



Extract the centerlines to get the vessel tree structure.

4. GRAPH REPRESENTATION



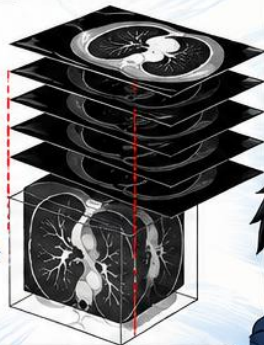
Convert the vessel tree into a graph of nodes and edges.

We convert the complex 3D vessel structure into a mathematical graph!

This helps us analyze structure, measure properties, and compare between subjects!

1 3D CT SCAN

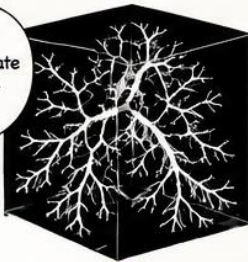
CT gives us cross-sectional slices of the lungs!



Stack many slices to build a 3D volume.

2 VESSEL ENHANCEMENT & SEGMENTATION

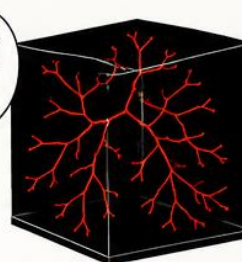
We enhance the vessels and separate them from other tissues!



Result: Binary mask of blood vessels.

3 SKELETONIZATION (MEDIAL AXIS)

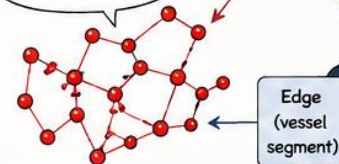
We thin the vessels to their centerlines while preserving the structure!



Result: 1D vessel tree (centerlines).

4 GRAPH REPRESENTATION

Each branching point becomes a node, and each segment becomes an edge!



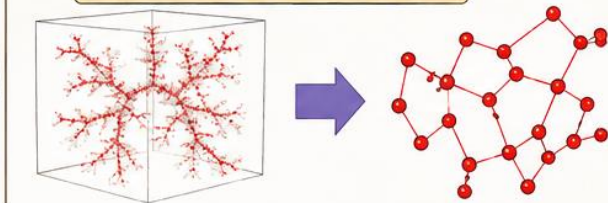
Now we have a graph that captures the topology of the vessel tree!

WHAT CAN WE DO WITH THIS GRAPH?

- ✓ Measure lengths, diameters, and tortuosity
- ✓ Analyze branching patterns and complexity
- ✓ Compare healthy vs. diseased lungs
- ✓ Use in simulations and machine learning

Graphs make complex structures easier to analyze and understand!

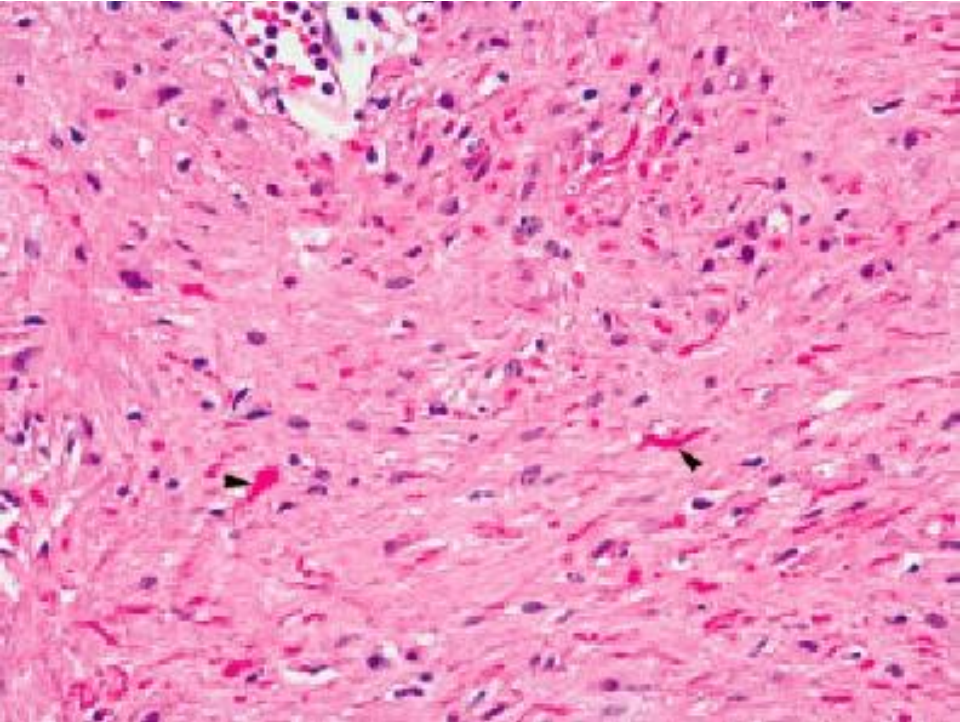
EXAMPLE OF FINAL RESULT



WHY IS THIS IMPORTANT?

This approach turns complicated 3D vascular structures into simple graphs, opening the door to powerful tools in mathematics, computer science, and medical research!

Optical Microscope



Rosenthal fiber: A thick, bright-pink, elongated structure seen under the microscope in brain tissue. It often looks worm-like, twisted, or corkscrew-shaped.



Albee Messing, Waisman Center

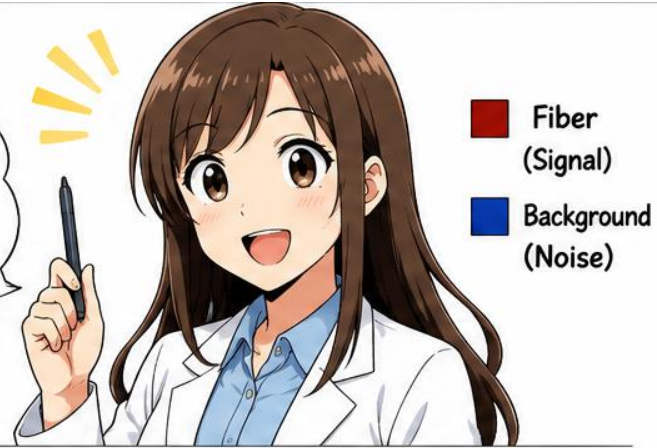
IMAGE SEGMENTATION (CLUSTERING)

Separating signal (fiber) from background (noise)



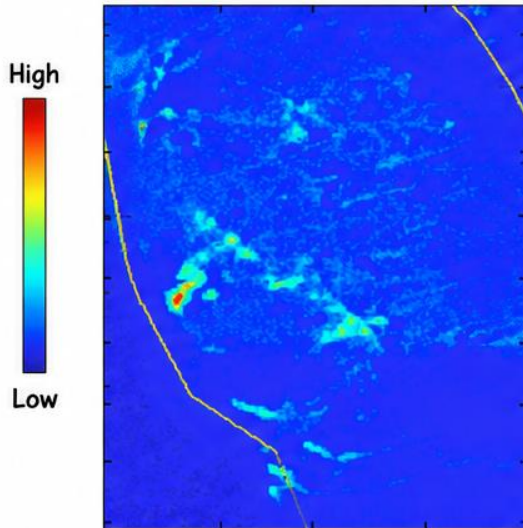
Image segmentation is the process of **partitioning** an image into different **tissue types** based on their **intensity** or **features**.

Our goal:
Detect the **fiber**
(**signal**)
and separate
it from the
background
(**noise**)!



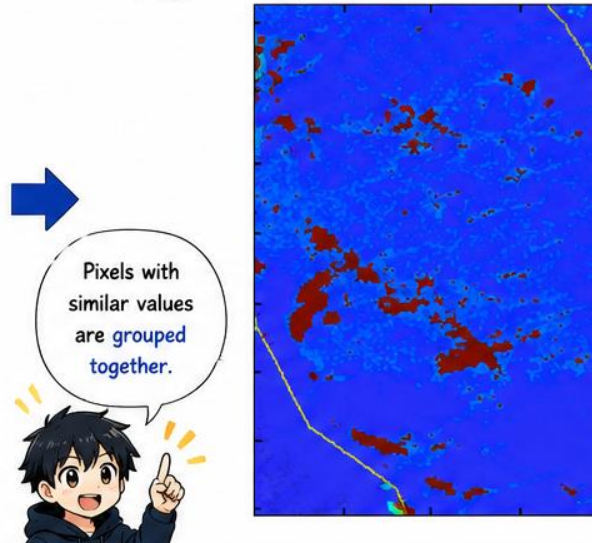
1 INPUT IMAGE (RAW DATA)

The image contains both the fiber (**signal**) and background (**noise**).



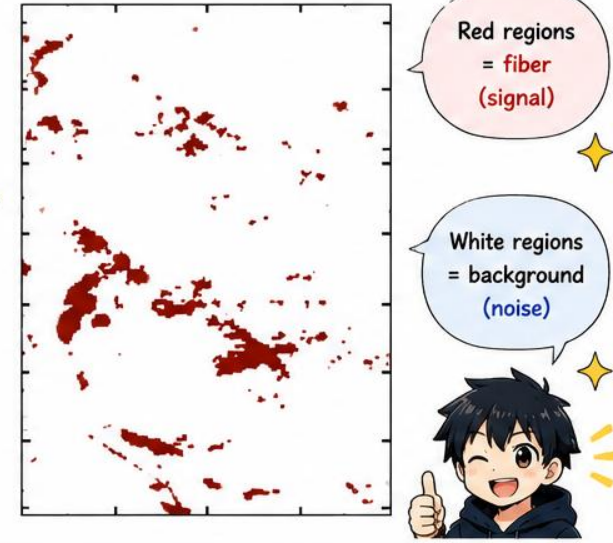
2 SEGMENTATION (CLUSTERING)

We group similar pixels (intensity/features) into clusters using a clustering algorithm (e.g., k-means).



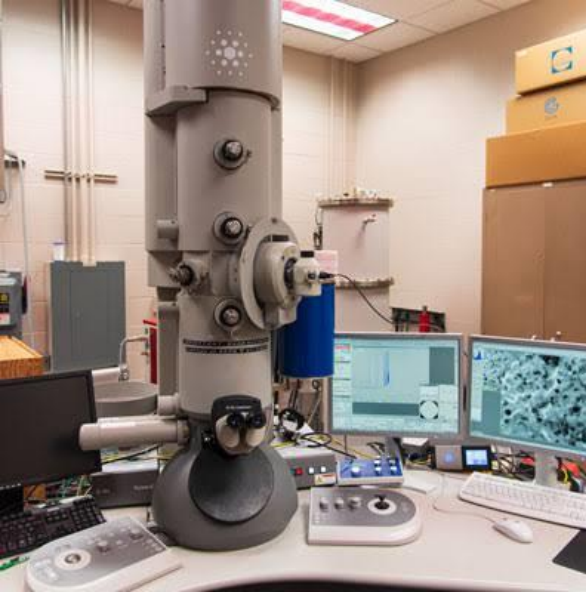
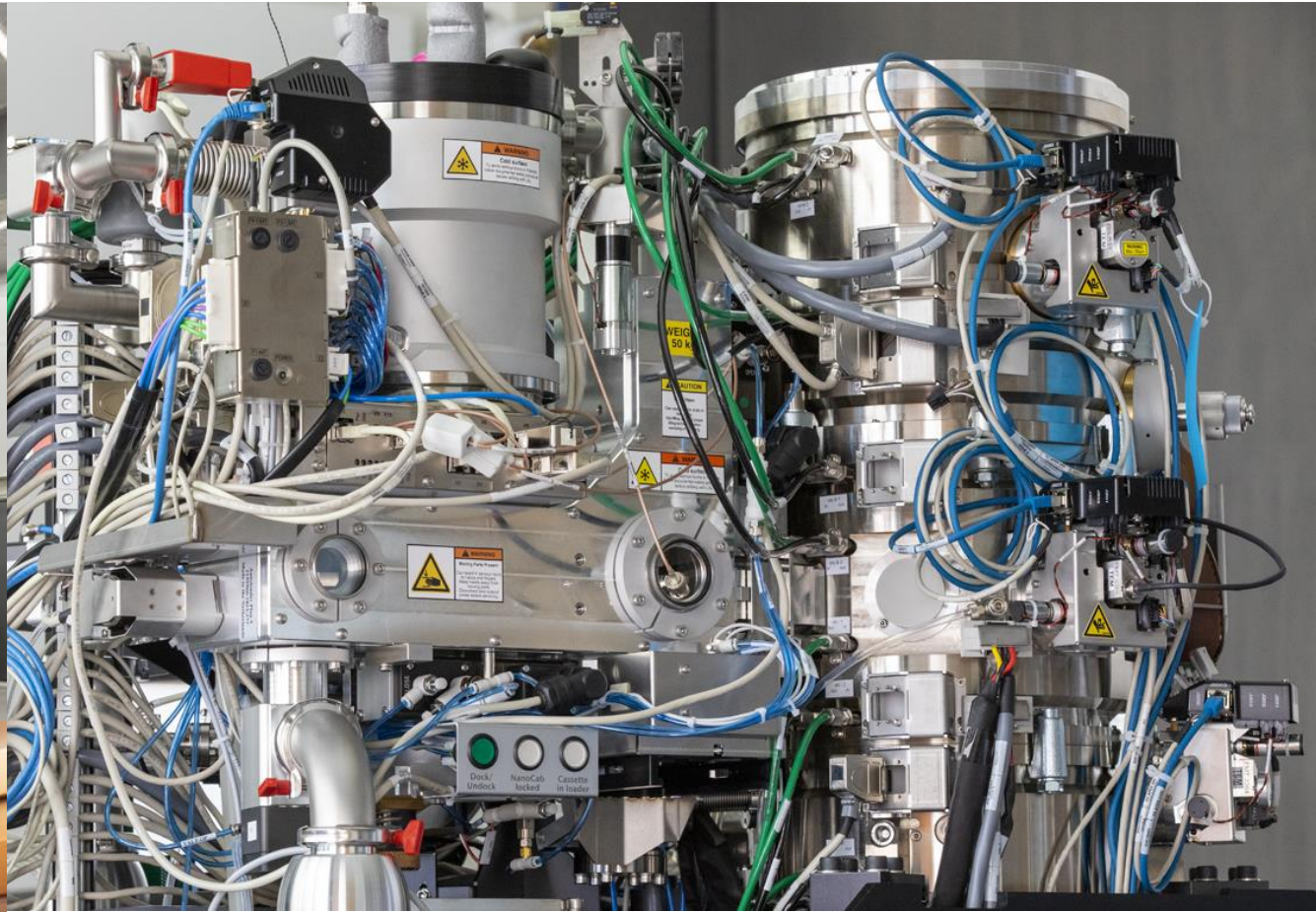
3 SEGMENTED RESULT

Fiber (**signal**) is separated from background (**noise**).



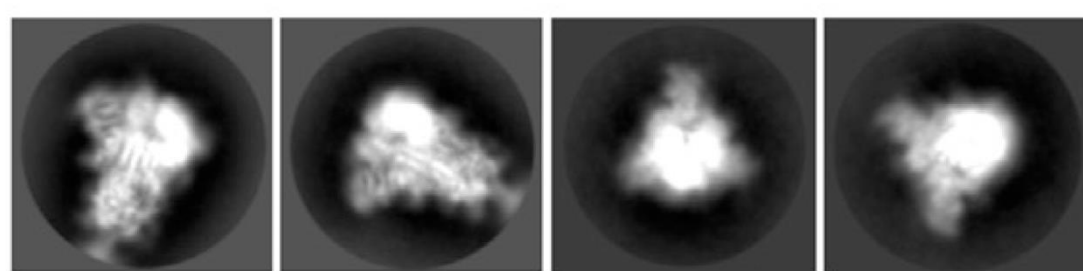
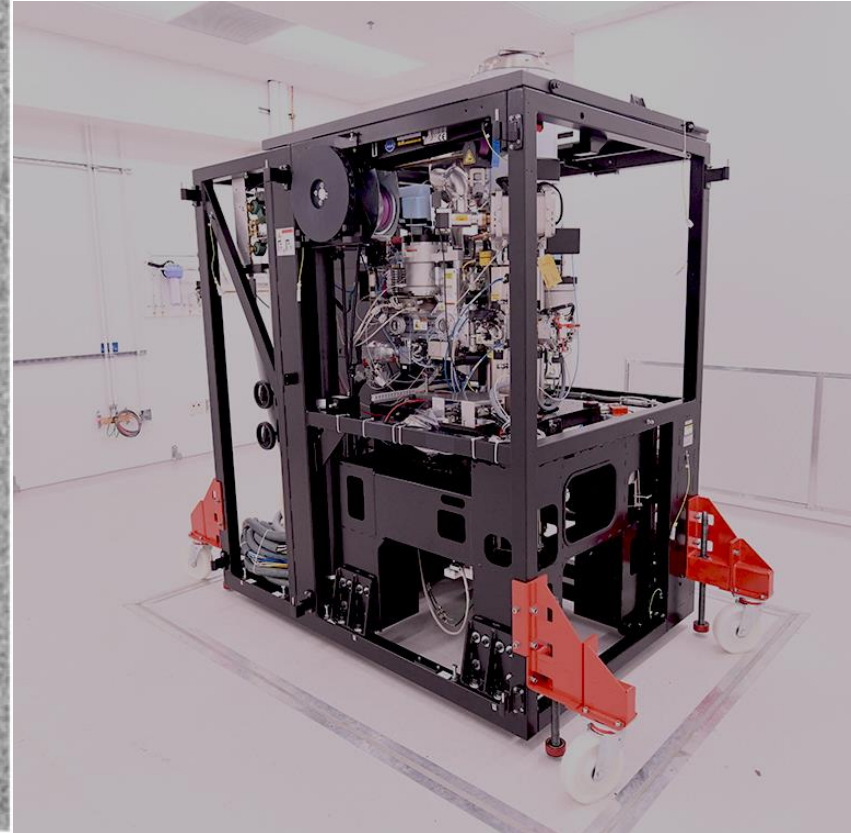
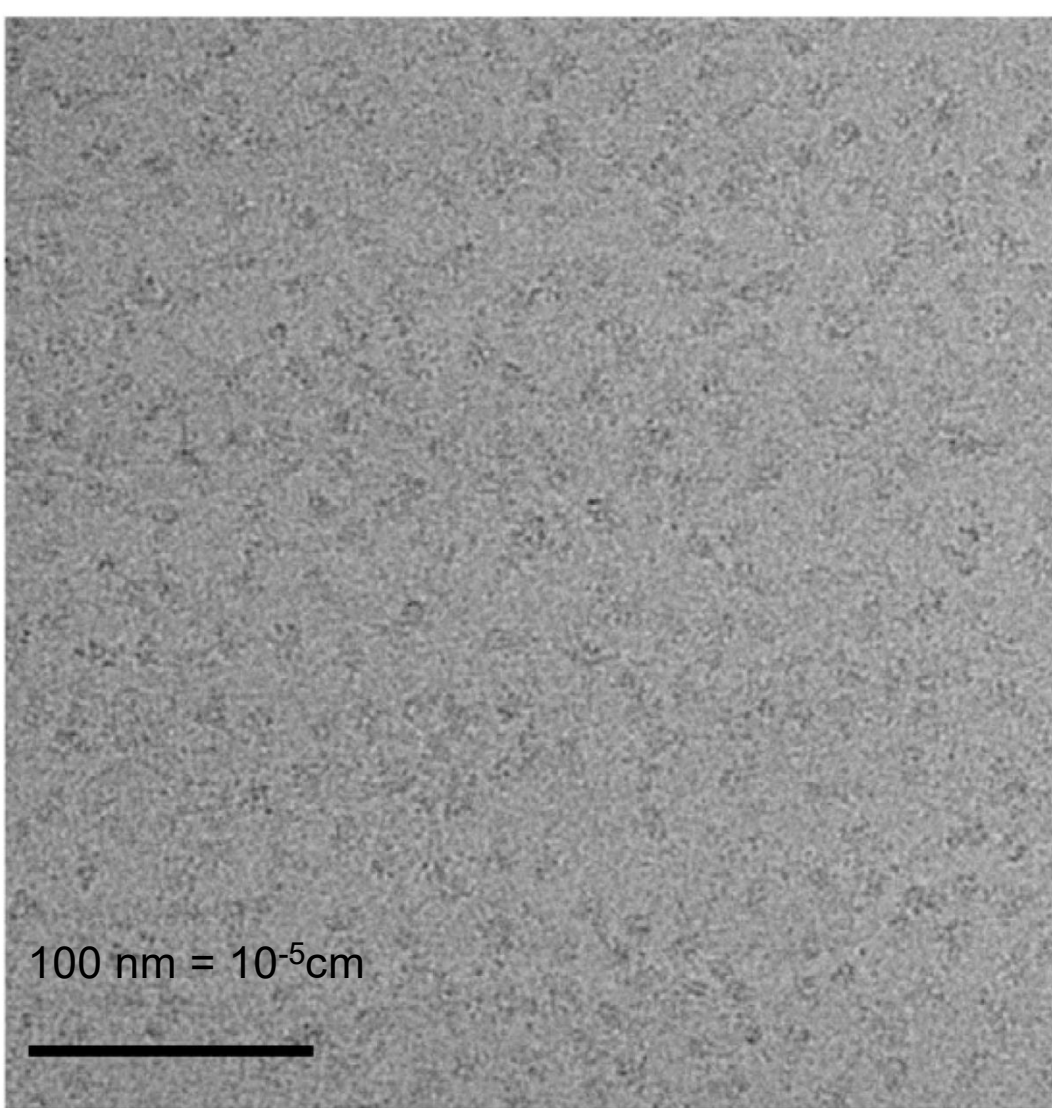
SEGMENTATION HELPS US FIND WHAT MATTERS BY REDUCING NOISE!

Cryogenic electron microscopy (Cryo-EM)



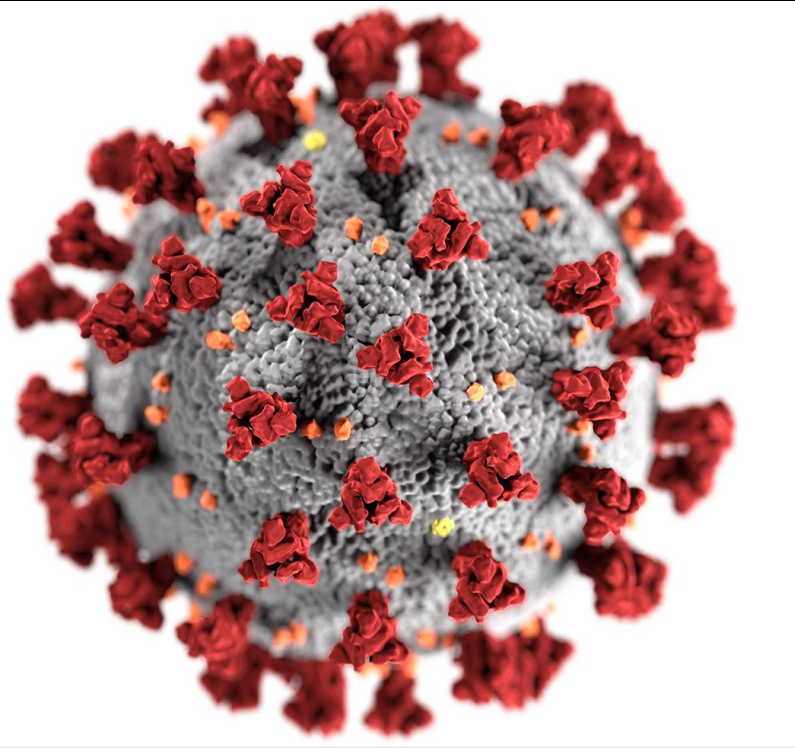
UW-Madison Cryo-EM Research Center

Cryo-EM Resolution: Freezing molecules

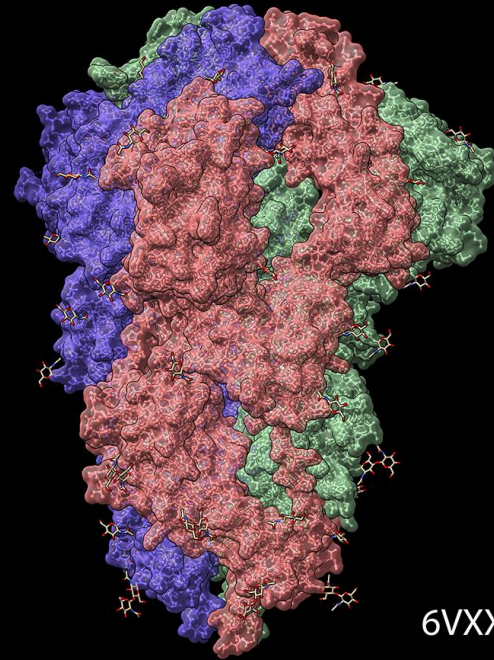


Spike protein image
From 300kV Titan Krios
Walls et al. 2020, Cell

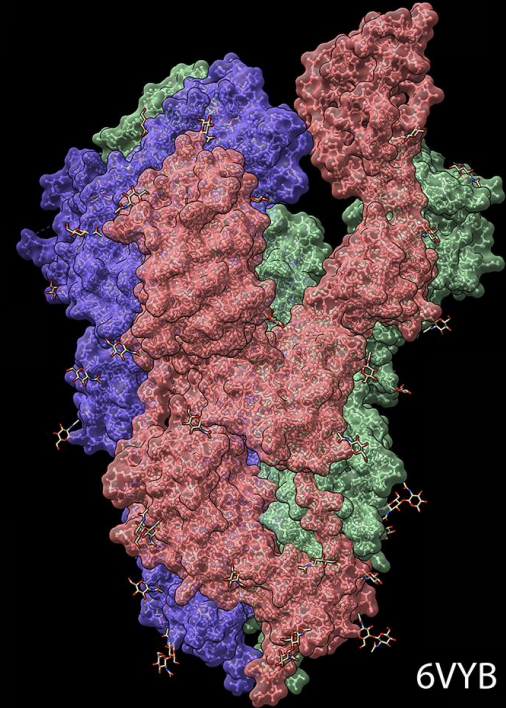
Conformational changes of spike protein



Covid-19 virus
Closed state

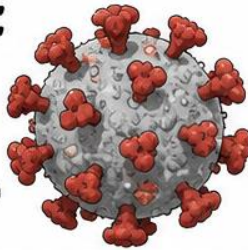


COVID-19 virus
Open state



CRYO-EM: SEEING LIFE IN NEAR-NATIVE 3D! ✨

How Cryo-Electron Microscopy works
Example: COVID-19 (SARS-CoV-2) Spike Protein



Cryo-EM lets us **visualize molecules** in 3D at near-atomic resolution—without crystals!



THE BIG IDEA

We freeze molecules in a thin layer of vitreous ice and use an electron microscope to record thousands of images. Computers then reconstruct the 3D structure!

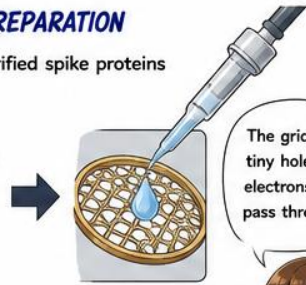
It's like taking thousands of tiny 2D photos and building a 3D model!

1 SAMPLE PREPARATION

We start with purified spike proteins from the virus.



Spike proteins in solution



A small drop is placed on a holey carbon grid.

2 VITRIFICATION (FLASH FREEZING)

The grid is rapidly plunged into liquid ethane, cooling it so fast that water becomes vitreous ice (non-crystalline).

Grid with sample



Vitrified grid

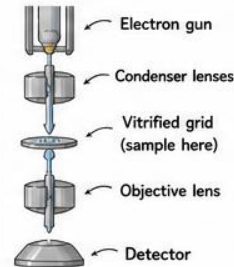


Liquid ethane (-183°C)

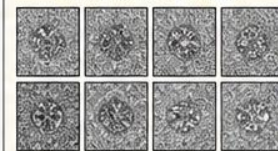
Vitrification preserves molecules in their near-native state without ice crystals!

3 IMAGING IN THE ELECTRON MICROSCOPE

The vitrified grid is placed in a high-vacuum electron microscope. An electron beam passes through the sample, and a detector records the image.



What we record: thousands of 2D images

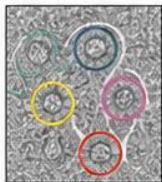


Each image = random orientation of a particle!

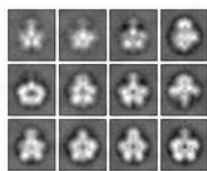
Each particle is in a **random** position and orientation.

4 FROM 2D IMAGES TO 3D RECONSTRUCTION

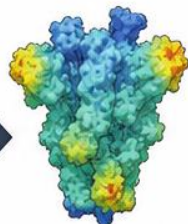
Computers pick out individual particles and group similar views together.



Picking particles



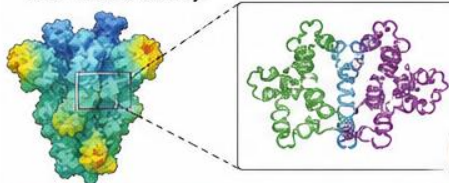
2D classification (many views)



Low density High density
3D reconstruction (density map)

5 INTERPRETATION & MODEL BUILDING

The density map reveals the shape. Researchers build an atomic model that fits the density.



✓ For the SARS-CoV-2 spike, cryo-EM revealed the prefusion structure, key for vaccine and drug design!

6 WHY CRYO-EM IS POWERFUL

- ✓ No crystals needed
- ✓ Works for large, flexible complexes
- ✓ Near-atomic resolution
- ✓ Captures multiple conformations
- ✓ Essential for understanding viruses, membrane proteins, and more!

Cryo-EM turns tiny, frozen samples into amazing 3D molecular pictures!

★ **CRYO-EM = Freeze it fast, image it many times, compute it smart, and reveal life in 3D!**



Unit Outcomes

- 1) Learned different image types - dimensionality
- 2) Can identify imaging data types
- 3) Load images to MATLAB
- 4) Visualized and rendered images
- 5) Get familiarized with images